# Carlos Romero Madrid

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**Computational Physicist • HPC Software Engineer • Creative Technologist**

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**To: The Selection Committee & Group Lead**
Data Analytics and Visualization Group
Barcelona Supercomputing Center (BSC-CNS)

## Subject: Application to join the Data Analytics and Visualization Group

Dear Members of the Data Analytics and Visualization Group,

Most resumes begin with a dense list of buzzwords. Let me begin with a sound: the hum of **MareNostrum 5** computing climate turbulence, or the chime of a motion-sensing sphere when my cat **Pipa** knocks it across the floor.

As a **Computational Physicist** (Double Degree Physics & Mathematics, 1st of Class MSc Plasma Physics) and **HPC Scientific Software Engineer** at BSC, I spend my days parallelizing solvers on Tier-0 supercomputers (MN5, Leonardo, LUMI). Yet my honest conviction is simple: *if petabytes of scientific discoveries remain locked behind cold terminal logs and abstract matrices, we have only completed half the mission.*

Inspired by **Domestic Data Streamers**, I treat data as raw physical matter to be touched, heard, and questioned. In my personal studio, I turn this into playful and rigorous reality:

- **MNusic (Sonifying & Visualizing MareNostrum 5 & LUMI):** Synaesthetic telemetry engine mapping cluster load, memory pipelines, and thermal gradients into real-time harmonic soundscapes (Web Audio API) and generative particles, letting audiences hear Europe's supercomputing heartbeat.
- **Pipa-Music-Ball (Bio-Kinetic Physical Computing):** Embedded interactive sphere (ESP32, 6-DoF IMU) translating feline kinetic play into microtonal scales. Proof that physical data interfaces can be both scientifically grounded and feline-approved.
- **Expo-Curator (Exhibition & Booklet Design Studio):** An open-source application for automated image sub-classification, catalog booklet design, and exhibition wall staging.
- **After Effects Lab (Procedural Motion Design):** Still photographic composition paired with motion graphics, procedural particle animations, and kinetic video in After Effects to turn static research figures into broadcast-ready visual essays.

I build at lightning speed: pairing with Large Language Models allows me to fluidly bridge C++/Fortran HPC kernels, microcontroller firmware, browser audio synthesis, and visual motion graphics.

Following your workshops and tracking your flagship initiatives—such as the urban digital twin modeling of **GDU-HPC**, the atmospheric narratives of **AIR**, and climate mobility in **REALLOCATE**—I see an organic match with my digital twin work at BSC (DestinationE / ClimateDT) and my creative installations. I would be thrilled to join Fernando Cucchietti and the team to co-design the next generation of daring, unforgettable scientific data experiences.

■ **Live Interactive Portfolio & Sound Installation:** I have built an interactive web version of this application featuring real-time MareNostrum 5 telemetry sonification, day/night audio crossfader, and video cinema mode at **https://carrommad.github.io/cool-cv**

Thank you for your time, consideration, and inspiration. I look forward to meeting and brainstorming upcoming projects.

Warm regards,

**Carlos Romero Madrid**

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